

RESEARCH STATEMENT

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Introduction and Motivation

My research lies at the intersection of graph theory and combinatorics, with a focus on *uniquely Hamiltonian graphs* — graphs that admit exactly one Hamiltonian cycle. The central open problem guiding my work is **Sheehan’s Conjecture** (1975): *there exist no uniquely Hamiltonian k -regular simple graphs for $k \geq 3$* . Despite decades of attention, the conjecture remains open for $k = 4$, and a general construction producing uniquely Hamiltonian 4-regular graphs has not been established.

A related conjecture of Sabidussi provides an important structural lens: *every Eulerian graph of minimum degree 4 admits a decomposition into cycles, none of which contains two consecutive edges of a given Eulerian circuit*. Fleischner’s 1994 construction [1] connects these two conjectures by transforming Eulerian circuits of $(4, 6)$ -regular multigraphs into dominating cycles and subsequently into unique Hamiltonian cycles. My research has focused on studying the limits of this approach.

Research to Date: Limits of Fleischner’s Construction

During my studies at the Department of Mathematics and Physics, Charles University, I investigated whether the construction due to Fleischner [1] can be extended beyond the Petersen graph to produce an infinite family of uniquely Hamiltonian 4-regular multigraphs. The construction proceeds by applying a sequence of graph operations on $(4, 6)$ -regular multigraphs with a fixed Eulerian circuit, with the aim of eliminating all *compatible cycles* — subgraphs whose edge-removal induces an Eulerian circuit and thereby compromises the uniqueness of the Hamiltonian cycle. My work identified obstructions to this construction, each of which gave rise to open questions.

Survival of compatible cycles. The operations in [1] target one compatible cycle at a time. I found that when a graph contains multiple vertex-disjoint compatible cycles, a single application destroys only the targeted cycle, leaving the others intact. Likewise, when two compatible cycles are *complements* of each other — their edge sets together covering all edges of the graph — one cycle is destroyed by the construction. This raises the question: whether the existence of a single Eulerian circuit is at all sufficient to guarantee a successful application of the construction, or whether a more refined characterisation of the circuit is needed.

Failure at large order. As graph order grows, so does the number of compatible cycles that survive the construction. To make this precise, I applied the construction to the $(4, 6)$ -regular graph derived from the **Blanuša snark** on 18 vertices (obtained by contracting the edges of a dominating cycle). The resulting graph retains compatible cycles that induce Eulerian circuits, confirming that the construction in [1] is not robust to increasing graph order. This naturally raises the question of classifying which graph classes obstruct the construction, and whether snarks form a complete such class.

The 4-regular obstruction. For 4-regular input graphs specifically, I showed that the construction produces a 3-regular graph in which the dominating cycle C_T coincides with the Hamiltonian cycle H_T , violating the uniqueness condition required by [1, Theorem 3]. This rules out 4-regular

graphs as valid inputs and points to a structural gap in extending the construction.

Cycle composition. A further obstruction is that the construction can generate new compatible cycles in the process of trying to eliminate existing ones. Specifically, when the construction is applied to two compatible cycles simultaneously, the resulting graph may acquire a new compatible cycle whose edge set is the union of the two originals. This suggests a deeper question about which transformations preserve or produce compatible cycles, and whether a formal characterisation of this phenomenon is possible.

The Petersen graph as a unique case. Taken together, these findings suggest that the Petersen graph is the *only* snark for which the construction in [1] succeeds, and motivate the following conjecture:

Conjecture. *The Petersen graph is the only snark that produces a uniquely Hamiltonian 4-regular multigraph through the Eulerian circuit formed by contracting edges not on its dominating cycle.*

Open Problems and Proposed PhD Research

The findings above lead to the following questions, which will form the core of my doctoral research:

1. **Characterisation of favourable Eulerian circuits.** Given a $(4, 6)$ -regular multigraph G with a fixed Eulerian circuit T , what structural properties of T guarantee that the construction in [1] yields no compatible cycle inducing an Eulerian circuit? Resolving this would directly address what makes the Petersen graph exceptional.
2. **Compatible cycle extension and Sabidussi's Conjecture.** I have shown that for $(4, 6)$ -regular graphs with at most one vertex of degree 6, a compatible cycle C can be extended to a compatible cycle decomposition equivalent to Sabidussi's Conjecture. The general case remains open: does every compatible cycle belong to some set S_0 , or can compatible cycles exist outside all such decompositions?
3. **Classification of obstructing graphs.** Which graph classes, beyond snarks, obstruct the construction in [1]? Does the construction hold in extremal cases for $(4, 6)$ -regular multigraphs?
4. **Cycle composition.** Under what conditions does the construction compose smaller compatible cycles into larger ones? A formal characterisation of this transformation may yield new invariants relevant to both the construction and to Sabidussi's Conjecture more broadly.

I am currently developing these questions with the goal of a publication on the scope and limits of the construction in [1].

Doctoral Goals

Determining whether a graph contains a Hamiltonian cycle is a well-known NP-complete problem, and while sufficient conditions for existence are established, no comprehensive characterisation of *uniquely* Hamiltonian graphs exists. My doctoral research will target precisely this gap, and the aforementioned open questions, focusing on Hamiltonian cycles in conjunction with Eulerian circuits in regular graphs.

My background spans both pure mathematics and computer science. Alongside research, I have

sustained a long-term commitment to mathematics education at secondary and university level, and I intend to continue contributing to teaching as part of my doctoral work.

References

- [1] H. J. Fleischner, *Uniqueness of maximal dominating cycles in 3-regular graphs and of Hamiltonian cycles in 4-regular graphs*, J. Graph Theory **18** (1994), no. 5, 449–459.
- [2] J. Sheehan, *The multiplicity of Hamiltonian circuits in a graph*, Recent Advances in Graph Theory, Proceedings of the Symposium held in Prague, 477–480, 1975.
- [3] H. Fleischner, *Eulerian Graphs and Related Topics*, Part 1, Vol. 1, Annals of Discrete Mathematics, Vol. 45, North-Holland, Amsterdam, 1990.